

Neural Texture Style Transfer on a 3D Mesh with Geometry Preservation

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1 Introduction

The goal of this project is to implement a neural style transfer method for a textured 3D mesh while preserving the original geometry of the object. Unlike ordinary 2D style transfer, where each image is stylized independently, this project optimizes the UV texture map of a 3D model. As a result, the stylized appearance becomes part of the object texture and remains visible from different camera viewpoints.

The selected task was:

Neural texture transfer on a 3D mesh with geometry preservation using UV map stylization.

The main idea is to keep the mesh geometry unchanged and optimize only the texture image attached to the UV coordinates. The 3D model is rendered from multiple viewpoints using differentiable rendering. The rendered images are then compared with a target artistic style image using neural style loss based on VGG19 feature activations.

In this project, a chair model from ShapeNet was used as the 3D object, and Edvard Munch's *The Scream* was used as the reference style image.

2 Project Objective

The main objective is to obtain a new UV texture map for a given 3D mesh so that the rendered object visually follows the target artistic style while preserving the recognizable structure and geometry of the original model.

The project includes the following components:

1. Loading a textured 3D mesh with UV coordinates.
2. Rendering the mesh from several camera viewpoints.
3. Optimizing the UV texture map through differentiable rendering.
4. Using a neural style loss based on Gram matrices.
5. Applying total variation regularization to reduce excessive noise.
6. Comparing the proposed UV-based method with naive 2D style transfer applied independently to rendered views.

3 Dataset and Input Data

The 3D model was taken from the ShapeNet dataset. The selected category was:

03001627 --- chair

The used model contained the following files:

File	Description
model_normalized.obj	Mesh geometry, faces, and UV coordinates
model_normalized.mtl	Material file linked to the texture
untitled/texture.jpg	Original texture image

The .obj file stores the mesh geometry, including vertices, faces, and UV coordinates. The .mtl file links the mesh with its texture map. In this project, the material file contained the following texture reference:

```
map_Kd untitled/texture.jpg
```

The original texture was a wooden texture, which made the chair appear as a wooden object before stylization.

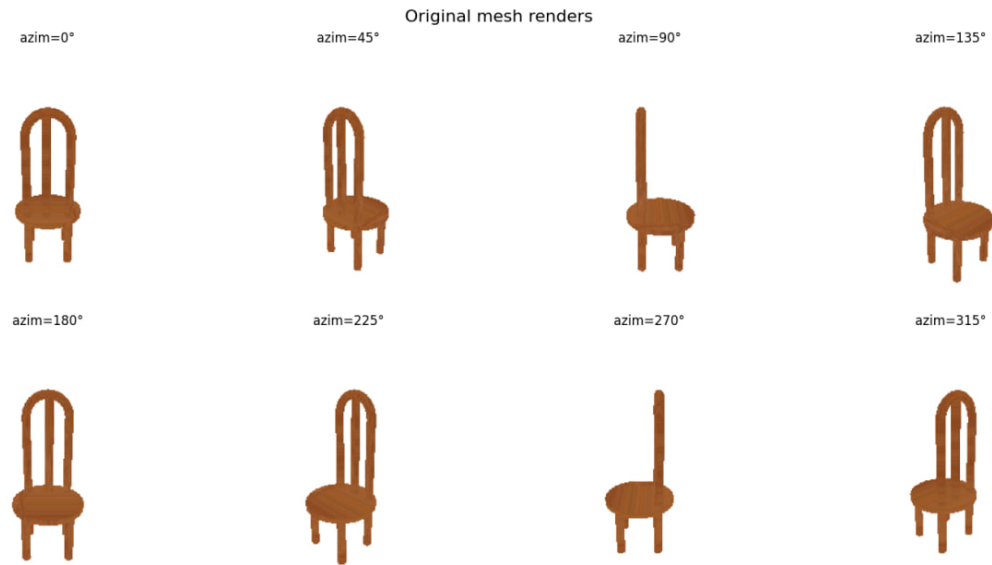


Figure 1: Original Mesh of chair

Original UV texture map



Figure 2: Original UV texture map

The style image was Edvard Munch's *The Scream*. This image was used as the artistic reference for the neural style transfer process.

Style image



Figure 3: Style image "Scream"

4 Methodology

4.1 Mesh Representation

The 3D object is represented as a textured mesh:

$$M = (V, F, U, T), \quad (1)$$

where:

- V is the set of vertices;
- F is the set of triangular faces;
- U is the set of UV coordinates;
- T is the UV texture map.

In this project, the geometry components V and F are fixed during optimization. Only the texture map T is updated.

Therefore, the optimization target is:

$$T^* = \arg \min_T \mathcal{L}(T). \quad (2)$$

The mesh geometry remains unchanged:

$$V = \text{const}, \quad F = \text{const}. \quad (3)$$

This ensures that the object remains recognizable as the same chair after stylization.

4.2 Differentiable Rendering

The mesh is rendered from different camera viewpoints using PyTorch3D. A differentiable renderer allows gradients to flow from the rendered image back to the texture map.

For a camera viewpoint c_i , the rendered image is:

$$I_i = R(M, T, c_i), \quad (4)$$

where:

- R is the differentiable renderer;
- M is the mesh;
- T is the current UV texture map;
- c_i is the camera viewpoint;
- I_i is the rendered image.

During optimization, several viewpoints are sampled so that the texture is optimized for multiple views, not only for one fixed camera.

4.3 Foreground Crop Using Alpha Mask

One issue in the initial implementation was that the object occupied only a small part of the rendered image, while most of the image was white background. If style loss is computed over the entire image, the VGG network mostly compares the white background with the style image, which weakens the style signal.

To solve this, the alpha channel from PyTorch3D rendering was used to crop the foreground object. The loss was then computed mostly on the object region instead of the full image.

Let A_i be the alpha mask of the rendered image I_i . The cropped foreground image is:

$$\hat{I}_i = \text{Crop}(I_i, A_i). \quad (5)$$

This improves the quality of optimization because the style loss focuses on the chair rather than the background.

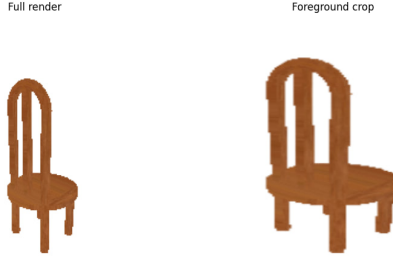


Figure 4: Foreground crop

4.4 Ambient-Only Lighting

The first versions used Phong-style lighting, where the rendered texture was affected by shadows and highlights. This reduced the visibility of the stylized texture. To avoid this, the final version used ambient-only lighting.

Ambient lighting helps display the texture colors more directly:

$$I_i = R_{\text{ambient}}(M, T, c_i). \quad (6)$$

This made the stylized colors more visible on the final rendered mesh.

5 Neural Style Transfer Loss

5.1 VGG19 Feature Extraction

The style loss is computed using a pretrained VGG19 network. The rendered image and the style image are passed through VGG19, and feature maps are extracted from several convolutional layers.

Let the feature map at layer l be:

$$F^l(I) \in R^{C_l \times H_l \times W_l}, \quad (7)$$

where:

- C_l is the number of channels;
- H_l and W_l are the spatial dimensions of the feature map.

The style layers used in the project were:

Style Layers

relu1_1
relu2_1
relu3_1
relu4_1
relu5_1

5.2 Gram Matrix

The style representation is based on Gram matrices. For a feature map F^l , the Gram matrix is computed as:

$$G_{ij}^l = \frac{1}{C_l H_l W_l} \sum_k F_{ik}^l F_{jk}^l. \quad (8)$$

The Gram matrix captures correlations between feature channels and represents texture and style information rather than exact spatial structure.

5.3 Style Loss

The style loss compares the Gram matrices of the rendered image and the style image:

$$\mathcal{L}_{style} = \sum_{l \in L} \left\| G^l(\hat{I}) - G^l(S) \right\|_2^2, \quad (9)$$

where:

- $\hat{I}$ is the cropped rendered image;
- S is the style image;
- $G^l(\cdot)$ is the Gram matrix at layer l ;
- L is the set of style layers.

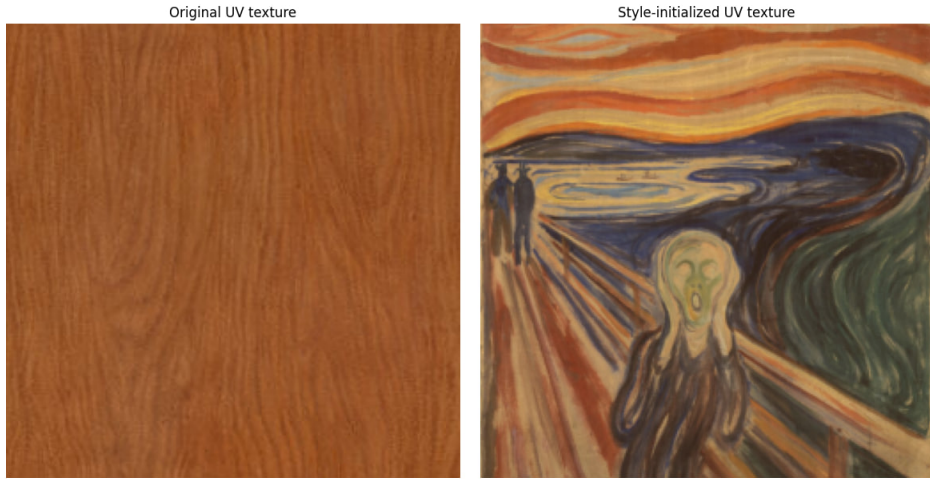


Figure 5: Original UV texture and Style-initialized UV texture

5.4 Total Variation Regularization

Total variation regularization was used to reduce excessive noise in the UV texture map. It encourages neighboring pixels in the texture to have similar values.

For a texture map T , the TV loss is:

$$\mathcal{L}_{TV} = \sum_{x,y} |T_{x+1,y} - T_{x,y}| + |T_{x,y+1} - T_{x,y}|. \quad (10)$$

This helps smooth the optimized texture.

5.5 Final Optimization Objective

In the final version, the loss function used style loss and TV regularization:

$$\mathcal{L} = \lambda_{style} \mathcal{L}_{style} + \lambda_{TV} \mathcal{L}_{TV}. \quad (11)$$

The final hyperparameters were:

$$\lambda_{style} = 1500, \quad (12)$$

$$\lambda_{TV} = 0.001. \quad (13)$$

Content loss was tested in earlier experiments but was removed in the final version because it strongly pushed the optimized texture back toward the original wooden texture. Removing content loss allowed the artistic style to become more visible.

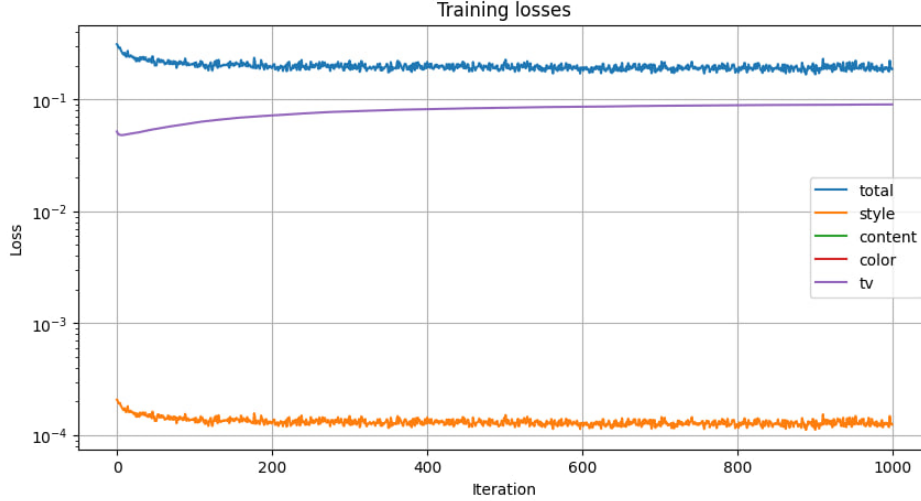


Figure 6: Loss graph

6 Texture Initialization

Initially, the UV texture was optimized starting from the original wooden texture. However, this produced weak stylization because the optimizer stayed close to the original appearance.

To make the style transfer more visible, the final experiment used a style-based initialization. The initial texture was created as a mixture of the original texture and the style image:

$$T_0 = 0.2T_{original} + 0.8T_{style}, \quad (14)$$

where:

- $T_{original}$ is the original wooden UV texture;
- T_{style} is the style image resized to the UV texture size.

This initialization made the optimization start closer to the target artistic style while still preserving some information from the original texture.

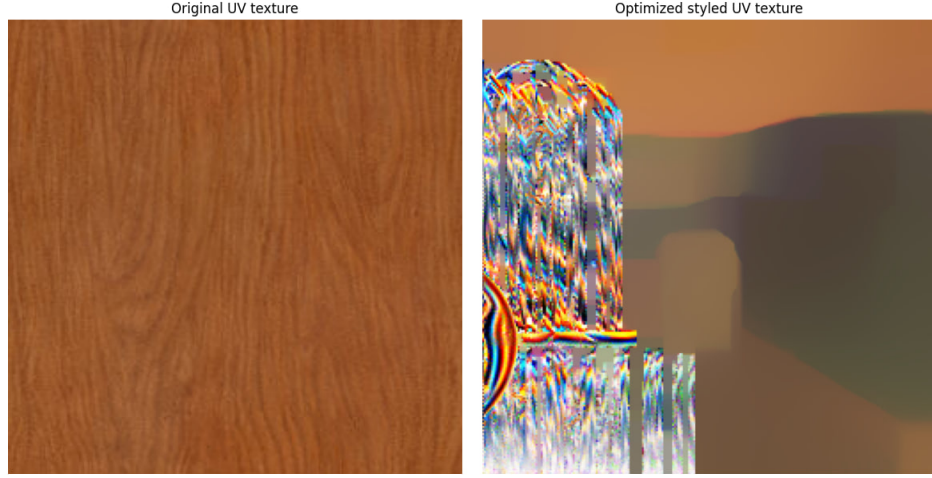


Figure 7: Original UV texture and Optimized styled UV texture

7 Viewpoint Sampling Strategy

A major challenge was that not all parts of the UV map receive gradients from a small number of camera views. For example, some parts of the chair such as the seat or legs may not be visible enough from random viewpoints.

To improve coverage, the final version used a mixed viewpoint strategy:

1. Random camera viewpoints during most iterations.
2. Fixed viewpoints every 8 iterations.

The fixed azimuth angles were:

$$\theta \in \{0^\circ, 45^\circ, 90^\circ, 135^\circ, 180^\circ, 225^\circ, 270^\circ, 315^\circ\}. \quad (15)$$

Higher elevation angles were also used to make the seat and upper surfaces more visible. This strategy helped optimize the texture across different visible parts of the chair.

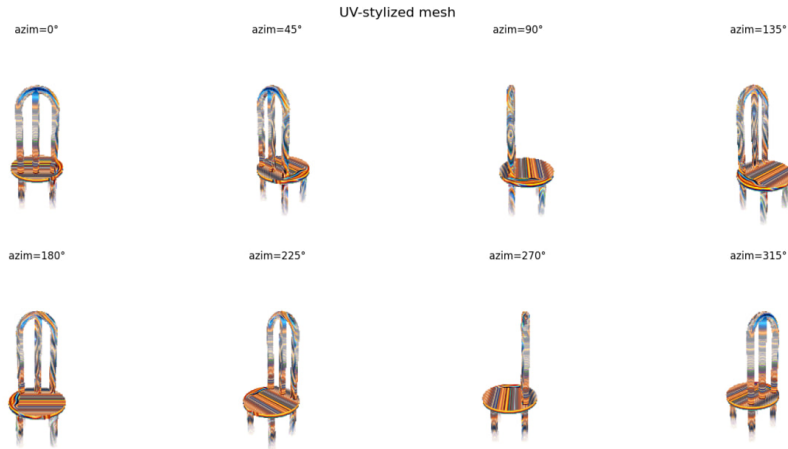


Figure 8: UV styled mesh

8 UV Coverage Map

A UV coverage map was computed to diagnose which pixels of the UV texture receive gradients during rendering.

The idea is to render the mesh from selected viewpoints, backpropagate through the rendered image, and observe which UV texture pixels receive nonzero gradients.

If a UV pixel receives gradient, it means that this part of the texture contributes to at least one rendered view.

This diagnostic step helps identify whether the stylization problem is caused by poor UV visibility.

The coverage map can be interpreted as:

$$C(u, v) > 0, \quad (16)$$

meaning that the UV pixel (u, v) is visible and receives gradient during optimization.

If a large part of the UV map has zero coverage, some texture regions cannot be effectively optimized from the selected camera views.

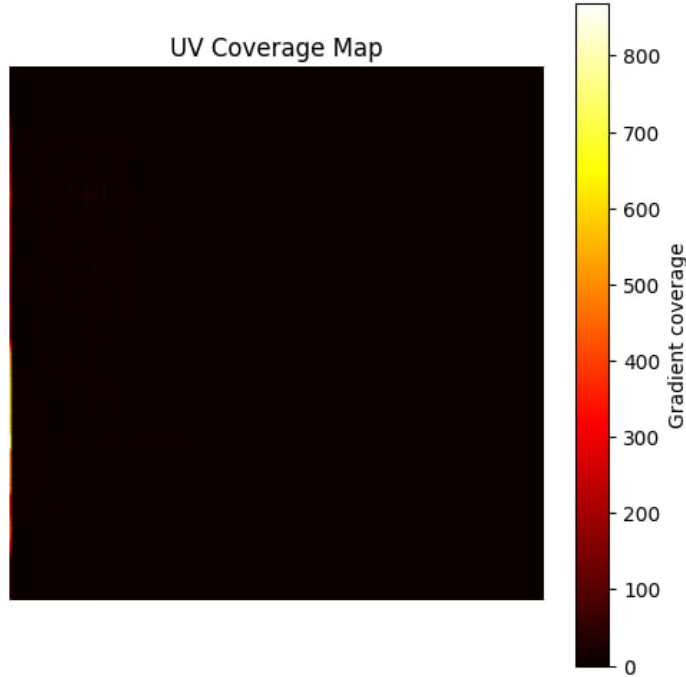


Figure 9: UV coverage map

9 Comparison with Naive 2D Style Transfer

To evaluate the UV-based method, it was compared with a naive 2D style transfer baseline.

9.1 Naive 2D Baseline

In the 2D baseline, the object is first rendered from several viewpoints. Then neural style transfer is applied independently to each rendered image.

This can be written as:

$$I_i^{2D} = \text{StyleTransfer}(R(M, T, c_i), S), \quad (17)$$

where each view i is stylized separately.

9.2 Difference from UV-Based Method

In the proposed UV-based method, only one texture map is optimized:

$$T^* = \arg \min_T \mathcal{L}(R(M, T, c_i), S). \quad (18)$$

Then this same texture is used for all viewpoints.

This is important because the UV-based method creates a real 3D-consistent texture. In contrast, the naive 2D method produces separate stylized images that are not consistent with each other in 3D space.

10 Experimental Results

The final comparison contained three rows:

1. Original rendered chair.
2. UV-stylized chair.
3. Naive 2D per-view stylized chair.

The original chair had a simple wooden texture. After UV-based stylization, the chair preserved its original shape, but the surface contained artistic patterns and colors inspired by the style image.

The 2D per-view style transfer produced stronger visual effects, but the style was applied independently for each viewpoint. Therefore, the patterns were not part of a single shared 3D texture. This means the result is not viewpoint-consistent.

The UV-based result was less aggressive than the 2D baseline, but it had an important advantage: the stylized appearance was stored in a single UV map and was consistent across different render views.

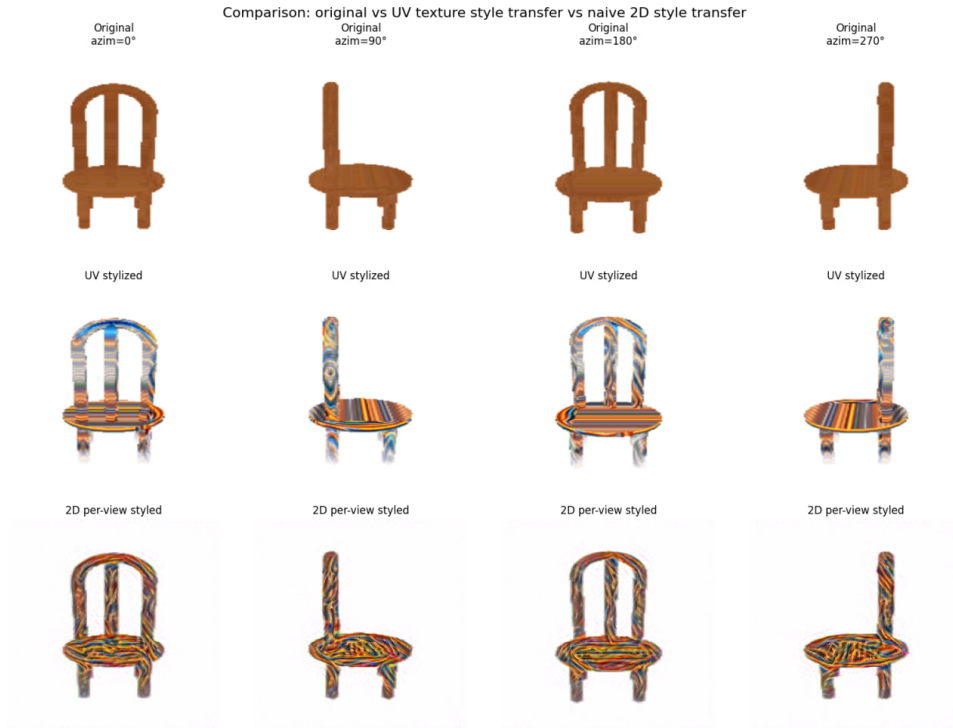


Figure 10: Comparison: original vs UV texture style transfer vs naive 2D style transfer

11 Discussion

The experiment showed that UV texture optimization through differentiable rendering can transfer artistic style to a 3D object while preserving geometry.

Several important observations were made.

First, computing style loss on the full rendered image was not effective because the white background dominated the image. Foreground cropping based on the alpha mask improved the optimization.

Second, Phong lighting reduced the visibility of the texture colors. Ambient-only lighting made the stylized texture more visible.

Third, content loss preserved the original wooden appearance too strongly. Removing content loss produced a more visible style transfer effect.

Fourth, viewpoint selection was important. Using only random views caused some parts of the UV map to receive weak gradients. The mixed strategy with fixed and random views improved coverage.

Finally, the UV-based result had some color artifacts and stripe-like patterns. This is expected because the UV layout of the ShapeNet model was not designed specifically for artistic texture transfer. Also, Gram matrix style loss captures texture statistics but does not preserve semantic structure from the style image.

12 Limitations

The project has several limitations:

1. The optimized texture contains visible artifacts and strong color stripes.
2. Some UV regions receive stronger gradients than others.
3. The final result depends on the UV layout of the original mesh.
4. Gram matrix style loss does not understand the semantic content of the style image.
5. The UV-based method is less visually aggressive than independent 2D style transfer.
6. The method requires differentiable rendering and is computationally more expensive than simple image-based stylization.

Despite these limitations, the main objective was achieved: the style was transferred to the UV texture map, and the geometry of the 3D mesh was preserved.

13 Conclusion

This project implemented neural texture style transfer on a 3D mesh using UV map optimization and differentiable rendering. A chair model from ShapeNet was rendered from multiple viewpoints, and its UV texture map was optimized using VGG19-based style loss and total variation regularization.

The final result demonstrated that the artistic style can be transferred onto the surface of a 3D object while preserving its geometry. The comparison with naive 2D style transfer showed that independent 2D stylization can produce stronger visual effects, but it lacks 3D consistency. In contrast, the UV-based method creates a single stylized texture map that remains consistent across viewpoints.

Overall, the project demonstrates the integration of differentiable rendering, optimization, and neural style transfer for 3D texture stylization.